PORTFOLIO PROJECT

E-Commerce Refund *Abuse Tracker*

Identifying high-risk customers and refund loss patterns using UCI Online Retail II dataset (2009–2011)

Pratham Singh · Excel · Pivot Tables · Risk Analytics

Problem & Objective

The Problem

Return abuse is a growing loss vector in e-commerce. Customers exploit liberal refund policies through:

- Excessive refund requests on valid orders
- Returning used or damaged items
- Multi-account exploitation for repeated abuse
- Coupon and promo misuse combined with returns

The Objective

Build a risk intelligence layer on top of raw transaction data to:

- Flag return-abusive customers automatically
- Quantify refund loss by customer, product, and month
- Classify customers into High / Medium / Low risk tiers
- Deliver an actionable dashboard for ops teams

Dataset & Methodology

Dataset

UCI Online Retail II

Rows	541,909 transactions
Period	2009 – 2011
Source	UK-based e-commerce
Returns	Invoice prefix 'C' + negative qty

Approach

1. **Flag returns**
   Identify cancellation invoices and negative quantities
2. **Engineer features**
   Return rate %, order value, repeat offender flag
3. **Score customers**
   High (>50% return rate), Medium (>25%), Low
4. **Aggregate insights**
   Pivot tables by month, product, customer, country
5. **Visualise**
   KPI cards + 4 charts with risk-based color coding

Key Findings

£143,445

Total Refund Loss

Across all return transactions

3,134

Total Returns

Cancellation invoices identified

445

High-Risk Customers

Return rate above 50%

18.4%

Avg Return Rate

Per unique customer

Top 2% of customers drive 38% of all refund value

A small cohort is responsible for disproportionate financial loss — a classic Pareto pattern in fraud.

Return abuse peaks in November–December

Seasonal spike aligns with promotional periods — suggest tightened return rules during sale events.

Top 10 abused products are low-value items

High-return products under £5 suggest policy exploitation rather than genuine quality complaints.

Dashboard Overview

Excel dashboard built with Pivot Tables, COUNTIFS, and conditional formatting. Add your dashboard screenshot below.

Risk Scoring Logic

High Risk

Threshold

Return rate > 50%

Excel Formula

```
=IF(Return_Rate_%>0.5,"High",...)
```

Recommended Action

Flag for manual review, restrict returns

445 customers

Medium Risk

Threshold

Return rate 25–50%

Excel Formula

```
...IF(Return_Rate_%>0.25,"Medium",...
)
```

Recommended Action

Monitor closely, soft restrictions

~1,200 customers

Low Risk

Threshold

Return rate < 25%

Excel Formula

```
...,"Low")
```

Recommended Action

No action required

~2,800 customers

Learnings & Next Steps

What I learned

1. Handling 540K+ rows with Excel performance optimization (subsampling + AGGREGATE)
2. Deduplicating customer-level metrics to avoid row-count inflation in KPIs
3. Identifying silent formula errors from blank Customer IDs in COUNTIF logic
4. Applying risk thresholds mirroring real-world fraud policy design
5. Structuring a dashboard for operational, not just analytical, use

Planned enhancements

SQL	Rebuild analysis in SQL for scalability across millions of rows
Power BI	Migrate dashboard to Power BI with DAX measures and live filters
Python	Add anomaly detection using pandas — flag impossible travel, device mismatches
ML	Experiment with logistic regression for predictive risk scoring

Thank you

Pratham Singh

pratham.singh2800@gmail.com

https://www.linkedin.com/in/prathamsingh9996/